LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Determination of CB size and maximum continuous current flow through CBs.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Determination of CB size and maximum continuous current flow through CBs.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of CB rating charts and wire gauge tables; Safety: Do not touch live circuits; Instructor supervision: Monitor trainee chart reading accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a Circuit Breaker in aircraft wiring?
(b) List two common causes of aircraft wiring degradation.
(c) Why is determining the correct CB size critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Circuit Breaker • Factors influencing CB selection • Relationship between wire size and current • Maximum continuous current flow limits • Standard CB sizing procedures	L & Ds	PPT, IB	15 min
(b)	• Calculation of load current • Selection of appropriate CB rating • Verification against wire ampacity • Impact of voltage drop on sizing • Review of standard sizing tables	L & Ds	PPT, PDF	20 min
(c)	• Common wiring degradation causes • Effect of overcurrent on insulation • Preventive maintenance measures • Troubleshooting degraded circuits • Documentation of CB replacements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wire size determines the maximum continuous current flow.
(b) Describe the procedure for selecting a CB size for a specific aircraft circuit.
(c) Identify three causes of wiring degradation related to current flow.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles learned in this lesson apply to the determination of required wire size and mechanical strength covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Determination of required wire size and mechanical strength.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min